Ghosts, Geometry, and Reality in Fourth-Order Gravity

A guided introduction to an AI-orchestrated research programme

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Abstract

Fourth-order gravity contains more local solutions than general relativity. Some enter familiar decompositions with an indefinite sign and are called ghosts. A sign is a serious warning, but it does not by itself identify a physical particle. Gauge reduction, global constraints, nonlinear continuation, boundary conditions, and quantum consistency are separate tests. This introduction explains an eighteen-paper programme that keeps those tests distinct.

The programme has not established that pure Weyl gravity is a viable theory of nature. It has established several narrower results. Positive and Krein completions of elementary fourth-order systems can be classified exactly, but their interaction continuations can fail. On compact gravitational laboratories, additional fourth-order directions can survive gauge reduction and have nonzero action-derived pairings, while nonlinear balance and resonance remove other directions. Strict pure Weyl gravity has a nonzero local Euclidean one-loop BV anomaly in the declared fixed-field-content complex; a compensator can make the tested cocycle exact only by changing the theory.

The newest results concern Schwarzschild black holes. Standard Lorentzian radiation conditions—future-horizon regularity and one-sided incoming or outgoing traces—do not select the Einstein subsector of the complete axial $\ell = 2$ pure-Weyl system. The repeated spin-two Regge–Wheeler block is a non-split self-extension. At one validated damped Schwarzschild frequency it has a length-two root chain: the geometric root is Einstein, whereas the generalized root has nonzero traceless-Ricci carrier. The compactly cut-off radial Green operator has a nonzero rank-one second-order pole, and the isolated local resonance contour contains the corresponding $te^{i\omega_n t}$ Einstein profile. This is not yet a global retarded ringdown theorem.

The repository is also a methodological experiment. A computer scientist without professional physics training orchestrates AI systems that propose, derive, challenge, verify, and document the research. The experiment can succeed only if experts can audit the assumptions, reproduce the evidence, and correct or reject the claims.

Authorship, accountability, and status

This is an open-repository preprint, not a peer-reviewed publication. GPT-5.6.sol is the principal programme author. Claude Fable 5 contributed to this introductory paper. Asger Alstrup Palm selected and coordinated the programme, made the public-release decisions, and is the accountable

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AI authorship is part of the experiment, not a substitute for evidence. Every strong claim is intended to be tied to a mathematical statement, declared assumptions, source code, a machine-readable certificate, and a verifier that can reject damaged inputs. The public repository is area9innovation/weyl-gravity. No repository-wide licence or archival DOI is claimed in this pre-release.

Contents

1 The ghost question is four questions

A nondegenerate fourth-order oscillator has an equation of the form

$$\left(\frac{d^2}{dt^2}+\omega_1^2\right)\left(\frac{d^2}{dt^2}+\omega_2^2\right)z=0.$$

It contains two second-order oscillators. In the standard Ostrogradsky description, one normally appears with the opposite Hamiltonian sign [1, 2, 3]. Bender and Mannheim showed that the unequal-frequency Pais–Uhlenbeck model also admits a non-Hermitian PT-symmetric description with a positive metric [4, 5]. At equal frequency, however, the two branches coalesce into a Jordan system.

These observations motivate the programme, but they do not settle gravity. The word "ghost" is often used for four different objects:

1. a local solution of a differential equation;
2. a classical direction remaining after gauge, constraints, charges, boundaries, and the symplectic radical are removed;
3. a direction that can be continued consistently through nonlinear interactions;
4. a state or particle in a positive quantum theory.

A calculation at one level is not automatically evidence at the next. In particular, a propagator residue is not yet a gauge-reduced state, and a classical flux sign is not yet a quantum probability.

Programme rule. Every result must name its theory, background, boundary conditions, phase space, and claim level. Results from different laboratories are compared only through an explicit map.

2 Why use several model universes?

No single background makes every part of the problem tractable. The programme therefore uses several declared laboratories.

Laboratory	Principal question
Pais–Uhlenbeck oscillator and scalar field	Positive metrics, Krein real forms, vacuum representations, Jordan limits, and interaction obstructions.
Conformal cylinder $\mathbb{R} \times S^3$	Complete free BV–BFV gauge complex, causal transport, and selected residual cohomology.
Berger and compact product backgrounds	Relational clocks, Einstein/additional wave comparison, action-derived pairings, nonlinear balance, and resonance.
Schwarzschild exterior and static spherical metrics	Horizon conditions, radiative traces, quasinormal resonances, Green poles, and static thermodynamic normalization.
Euclidean backgrounds and reduced mode spaces	Local anomaly classes, determinant data, and candidate two-point functions.

The laboratories are not asserted to be one physical universe. They isolate different promotion gates. A compact positive block does not establish an asymptotically flat particle, and a black-hole endpoint theorem does not establish a cosmology.

3 What the first six papers establish

Papers 01–03 reconstruct the elementary fourth-order completion problem. They classify the canonical positive symplectic diagonalization of the unequal-frequency Pais–Uhlenbeck system, identify a minimum-distortion metric, and distinguish metric geometry from vacuum and Fock-sector geometry. Their common lesson is that "the metric" is not one universal object: a finite oscillator similarity transform, a field representation, and a vacuum covariance have different continuity and ultraviolet requirements.

Paper 04 carries that distinction into linearized Einstein–Weyl gravity. Gauge reduction removes redundant variables, but it does not turn a fourth-order gravitational system into five independent copies of a scalar oscillator. Lorentz covariance, helicity, constraints, and the conformal critical limit constrain which real forms are admissible.

Papers 05–06 ask whether a free positive completion survives interaction. Their scoped answer is negative for the canonical analytic continuation tested there. Resonant branch conversion obstructs the required metric deformation in the scalar field and in an open Einstein–Weyl gravitational channel. This does not exclude every indefinite, nonlocal, singular, or nonperturbative prescription. It shows that solving the free theory is not the same as solving its interaction problem.

4 Causal reduction and compact classical tests

4.1 Residual cohomology is not a graviton count

Papers 07–08 analyze free pure Weyl gravity on the conformal cylinder. One paper computes a selected residual complex after treating all fifteen residual conformal generators as constraints in a declared zero-charge sector. The first surviving classes are

$$H^4_{\rm res} = \mathrm{span}\{[W_+^2], [W_-^2]\}, \qquad G_{\rm res} = I_2.$$

They are centered deformation or vertex classes, not one-particle gravitons. The companion causal paper constructs the covariant free BV–BFV transport that connects compact and global support conditions and transports the relevant pairings. Neither paper constructs a positive interacting graviton Hilbert space.

4.2 Additional classical directions can be real

Papers 09–13 move to clock and compact Einstein–Maxwell/Weyl–Maxwell laboratories. They show why local solution counting is insufficient in both directions:

- some additional axial and polar fourth-order directions survive local gauge reduction and have nonzero Lee–Wald pairings;
- the sign pattern is not simply "Einstein positive, additional negative";
- at second order, formal finite-harmonic continuation is controlled by five global stabilizer-charge conditions;
- bounded continuation has further secular-growth and resonance conditions;
- a causal changed gravity–clock parent can still fail the physical health test through a Hamiltonian–Hopf quartet.

The resulting Phase-1 verdict, synthesized in Paper 15, is a classification. It neither rescues pure Weyl gravity nor proves a universal no-go theorem. The programme found real additional classical directions and real obstructions, but no tested candidate reached a positive full-BV quantum state, scattering theory, or unitarity theorem.

5 The Schwarzschild result

5.1 Radiation conditions do not select the Einstein sector

Maldacena-type boundary selection can isolate an Einstein sector of conformal gravity in Euclidean anti-de Sitter or late-time de Sitter settings [6]. Paper 16 asks a different question: do the standard Lorentzian black-hole conditions—future-horizon regularity together with incoming, outgoing, or no-incoming traces—force the additional pure-Weyl curvature carrier to vanish?

For the complete axial $\ell = 2$ Schwarzschild system, the answer is no. The radial differential module has a filtration with successive factors

$$M_{\mathrm{RW}}^{(2)}, \qquad M_{\mathrm{RW}}^{(2)}, \qquad M_{\mathrm{RW}}^{(1)}.$$

The two spin-two factors are a non-split self-extension, not a direct sum. The complete incoming trace space has an action-derived nondegenerate form of signature $(1, 2)$. Exact scalar Wronskian arguments show that every incoming direction is populated by a future-horizon-regular exterior solution for every real $\omega > 0$. Outgoing population is certified on a band and holds away from isolated reflection zeros. The same filtration excludes exponentially growing separated axial modes.

These statements coexist without contradiction. Separated-mode stability means that the tested system has no exponentially growing separated frequency. It does not make the inherited flux positive and does not bound the full time-domain resolvent.

5.2 A defective quasinormal resonance

At one validated simple Schwarzschild spin-two quasinormal frequency, the extension selector is nonzero and the spin-one factor is a local unit. The complete connection then has Smith valuations

$$(0, 0, 2).$$

Its algebraic multiplicity is two, its geometric multiplicity is one, and it has a length-two root chain. The geometric root lies in the Einstein submodule; the generalized root has nonzero gauge-invariant traceless-Ricci carrier.

Paper 17 identifies the underlying projective class:

$$[\mathcal{I}_{\mathrm{Bach}}] = \frac{i\omega}{2}\left[1 - \frac{2}{r}\right].$$

Writing $m = \mu^2$ for the signed squared-mass parameter, the complete coupled massive axial system has tensor projective class

$$[\mathcal{I}_{\mathrm{phys}}] = \frac{1}{3}\left[1 - \frac{2}{r}\right], \qquad [\mathcal{I}_{\mathrm{Bach}}] = \frac{3i\omega}{2}[\mathcal{I}_{\mathrm{phys}}].$$

Thus the identification is an exact first-jet statement for the complete coupled equations, not merely for a scalar graded quotient. An all-orders differentiated-Jost construction is still needed before converting the intrinsic selector into a physical massive-QNM slope. The exact local crosswalk gives the generalized root the same critical-confluence interpretation as the logarithmic partners of critical gravity [7, 8], but the asymptotically flat endpoint tangent is polynomial rather than a literal independent radial $\log r$ mode.

The finite-radius radial Green problem, and equivalently its compactly cut-off exterior realization, has a nonzero rank-one second-order pole:

$$R(\omega) = \frac{R_{-2}}{(\omega - \omega_n)^2} + \frac{R_{-1}}{\omega - \omega_n} + R_{\mathrm{hol}}(\omega), \qquad R_{-2} \neq 0.$$

Exact axial reconstruction shows that the Einstein range of R_{-2} has nonzero Bondi shear. A complexified conserved, traceless, radially compact odd source can have nonzero adjoint overlap. On an isolated local resonance contour, the contribution therefore contains

$$e^{i\omega_n t}\big(V_1 + it\, V_0\big),$$

where V_0 is the ordinary Einstein QNM and V_1 is the generalized, non-Einstein partner.

Exact boundary of the ringdown claim. The local cut-off radial Green pole and the isolated-contour $te^{i\omega_n t}$ term are established. A global causal spacetime resolvent, inverse-Laplace contour deformation, complete late-time QNM expansion, and real astrophysical source calculation are not.

5.3 Static black holes are a separate problem

Paper 18 separates the static spherical Bach-flat classification and horizon normalization problem from the perturbation theorem. Within the declared Laurent class it proves the corrected Einstein sheet and shows that the bare chart-normalized Iyer–Wald charge is nonintegrable. Requiring the field-dependent generator to descend to the residual quotient forces $\chi = uf(J)\partial_t$. The resulting exact Hamiltonian, Wald entropy, and signed temperature satisfy one first law simultaneously at every simple horizon. The theorem is static and linear-spherical at charge level; it is not a physical-process or radiative thermodynamics theorem.

6 The quantum fork

Paper 12 computes a nonzero local Euclidean one-loop Weyl-anomaly class for strict fixed-field-content pure Weyl gravity in its declared BV complex. A formal compensator makes the tested cocycle exact, but enlarges the field content and counterterm space. It is therefore a result about a changed theory, not proof that strict pure Weyl gravity is anomaly-free.

A separate reduced free-wave construction admits Hadamard two-point distributions with one positive and two negative Krein signs. This is a well-defined reduced carrier, not a positive full-BV state. The repository does not contain a complete Lorentzian off-shell BV propagator, BRST-compatible full-metric Hadamard state, renormalized time-ordered products, Lorentzian quantum master equation, physical particle space, or unitarity theorem.

7 What has been learned

Five conclusions survive the changes of model and method:

1. **Pole signs are warnings, not complete physical verdicts.** The final object must be named: local solution, reduced classical direction, nonlinear tangent, or quantum state.

2. **A direct-sum factorization can hide an extension.** In Schwarzschild pure Weyl gravity the repeated spin-two factor is non-split, and the extension becomes spectrally visible at a defective QNM.

3. **Radiation conditions select propagation direction, not extension layer.** Standard Lorentzian endpoint conditions do not impose the additional traceless-Ricci condition defining the Einstein subsector.

4. **Stability, sign, and observability are different tests.** The axial system can lack exponentially growing separated modes while retaining an indefinite populated flux space and a double-pole response.

5. **Changing the theory must be stated.** Compensators, altered boundary conditions, and different positive metrics can be useful, but they are not silent proofs about strict pure Weyl gravity.

8 The AI-orchestrated research experiment

Asger Alstrup Palm is a computer scientist, not a professional physicist. He uses AI systems as a research team: to propose questions, derive identities, implement symbolic and interval computations, search the literature, conduct adversarial reviews, and draft papers. Palm chooses the programme and decides what is released.

This arrangement creates obvious risks. AI systems can produce plausible but false derivations, reinforce one another's mistakes, invent references, or conceal uncertainty behind polished prose. The repository therefore treats the workflow itself as an experiment. Its governing rules are:

- fail closed when data, assumptions, or verification are missing;
- preserve failed and superseded results in append-only records;
- distinguish reproduction from independent verification;
- distinguish exact algebra from validated or exploratory numerics;
- state what a calculation does not establish;
- invite expert criticism at the level of equations and certificates.

The repository cannot validate itself by declaring the experiment a success. Scientific success requires outside experts to reproduce, criticize, correct, cite, or use the work. A decisive reproducible refutation would also count as useful progress.

9 Paper map

Paper	Layer	Main subject
01–03	Free systems	Positive metrics, minimum distortion, vacuum representations, and the Krein/Jordan boundary.
04–06	Gravity and interaction	Gauge reduction, Einstein–Weyl completions, and interaction obstructions.
07–08	Causal residual theory	Pure-Weyl residual cohomology and covariant BV–BFV transport on the conformal cylinder.
09–13	Compact laboratories	Clocks, Einstein/additional waves, gravity–light transfer, anomaly, and nonlinear tangent cones.
14	Black-hole programme	Initial pure-Weyl black-hole radiation architecture; superseded claims are corrected by later papers.
15	Synthesis	Four-level Phase-1 classification and nonpromotions.
16	Endpoint theorem	Standard Lorentzian endpoint conditions do not select the Einstein subsector.
17	Resonance theorem	Non-split axial $\ell = 2$ self-extension, defective QNM, and cut-off exterior Green double pole.
18	Static charges	Residual-basic normalization and simultaneous signed first laws at every simple horizon of the certified static family.

10 What remains open

The most important next steps are now narrower than the original ghost question:

1. construct a global causal Schwarzschild operator domain compatible with the differentiated outgoing state, and justify the retarded contour deformation;
2. promote the exact complete coupled massive axial first-jet crosswalk to all-orders analytic differentiated Jost maps;
3. compute real source excitation and asymptotic detector coefficients;
4. finish the polar and broader multipole black-hole analyses;
5. determine whether any changed classical theory has an open, physically healthy parameter region rather than an isolated causal fixture;
6. construct, or rule out under explicit hypotheses, a positive BRST-compatible Lorentzian quantum state and interacting observable algebra;
7. subject the central certificates and papers to independent expert review and archival reproduction.

11 Conclusion

The programme's present conclusion is deliberately modest. Pure Weyl gravity has additional classical structures that cannot all be dismissed as gauge, duplicated factors, or bad horizon behavior. It also has serious classical, nonlinear, boundary, and quantum obstructions. The most developed black-hole result is structurally sharp: the axial Schwarzschild system contains a non-split Regge–Wheeler self-extension, and one ordinary Schwarzschild QNM becomes a defective pure-Weyl resonance with a nonzero double radial Green pole.

That is scientifically interesting without being a theory of observed gravity. The remaining tasks—global causal propagation, real excitation, complete asymptotics, full gauge reduction, and quantum positivity—are exactly the tasks that determine whether the structure is physical.

Current verdict. The ghost problem has not been solved. It has been decomposed into falsifiable tests, several of which now have exact answers. No tested theory in the programme has yet passed all four levels.

Repository and verification. Sources, committed PDFs, computational supplements, certificates, and verifiers are available at github.com/area9innovation/weyl-gravity. The most stable Phase-1 account is Paper 15; Papers 16 and 17 contain the current black-hole endpoint and resonance results. Historical records are retained even when later work narrows or supersedes their interpretations.

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